Embedded Systems International

Project Statement of Work (PD3)

Team Identifier: S4-B

Team Name (optional):

Team Member Names: Marina Mannan, Prakarsha Poudel, Preston Gaskill, Spencer Opitz

Submit your document as a PDF file in Canvas under the corresponding project assignment.

One of your section's lab TAs must approve of this Statement of Work by adding a comment in the Canvas assignment. The team is responsible for requesting approval promptly after submission.

Refer to the Project Requirements document before completing this Statement of Work (SOW). A statement of work is a focused concise proposal and agreement that describes work to be done. Teams should complete and submit this SOW form, which represents several parts of a statement of work, including a plan for what you are doing and how. The SOW defines the scope of your project and the approach you are taking to deliver on the goals.

Problem Statement

First, has your team reached consensus on the autonomous vehicle (AV) application you will use as the context or story for your project? All projects, regardless of application, will need to meet the same basic requirements and will be recognized for innovative features.

To define your problem, think about one or more users and their needs. Write at least one Point of View (POV) statement for your application. Follow Steps 2 and 3 in the following guide (also in Canvas as a PDF document):
Define and Frame Your Design Challenge (links to IDF page) (URL: https://www.interaction-design.org/literature/article/define-and-frame-your-design-challenge-by-creating-your-point-of-view-and-ask-how-might-we)

Next, think about and write a few sentences that give a high-level summary of the broad mission goals for your AV application. For example:
The purpose of the AV is to provide … .
The AV will be capable of doing … .

The mission goals and user needs establish the purpose of the project and why you are working on it. Now you should translate these into a more detailed problem statement that provides a specific, concise, clear and thorough description of the context for the problem, an explanation of user needs that will be addressed, and an outline of your proposed technical approach to solving the problem.

Problem Statement

POV statement:
-Amazon Drop off robot
The reason why is something they waste time by waiting for someone to receive it, some people have pets outside which scares the drivers to deliver, delaying the time to deliver. The weather can also be a factor, like rain can make it difficult for the person to drop off properly. (basically drops off package from road to the doorway home)
Delivery driver gives package to robot from the road and robot puts it on the doorway of the house while avoiding trees(tall obj), rocks(small obj) and cliffs(pits) in the garden etc…
-Boundary is white tape equivalent to not getting outside the garden, on the road.
-Package on top of the cybot(could be a small object)
-Make it completely manual controlled (easier this way)
- We could have a button, say 'm', which when pressed calls a function to play music.
-Keep moving straight until u detect white tapes, and if we do, back up, change to face left or right.
-While detecting while tapes and moving straight, if bumped onto a short object, backup and navigate accordingly
-Keep looking at the Putty data values for tall objects and make sure we don't hit them at all.
-180 degree scan can be done extremely quickly(that's what the TA told me), will ask him how.

High-level mission goals:
Efficient package delivery: Allow Amazon delivery drivers to drop off packages with a remotely controlled Delivery Robot stationed in the garden.
Avoid obstacles: Navigate around trees, rocks and possible cliffs in the garden.
Travel to the doorstep: The doorstep is indicated by four thin pillars arranged in a square fashion. When arrived, let the owner know we're here by playing a sound.

Short description of the problem: Amazon delivery drivers encounter everyday obstacles like adverse weather, problematic pets, and unresponsive recipients, all slowing down their deliveries. Making deliveries through a Package Delivery Robot streamlines this process. Drivers can leave packages with

the robot in the garden, then the robot will deliver it to the doorstep and play a short music upon arrival to let the recipient know their package is here, ensuring efficient and reliable delivery, regardless of external factors.

In addition to writing a paragraph about the problem, you are to draw a **problem sketch: a one-page sketch illustrating your solution with a user context (big picture view)**. This should show the scope of your work in relation to one or more user needs. Refer to the sample project sketches.

Design Approach

Next, consider your AV application in relation to the project requirements and the five categories by which it will be evaluated.

1) Functionality in relation to the AV application mission goals and user needs
2) Mapping of functional requirements to platform components and capabilities
3) Elements of the test field
4) Serious incident penalties
5) Feature bonuses

In this section, you will identify and describe how you will design your application for each of these categories.

To complete the tables below, your team may want to use ideation tools, such as Lotus Blossom. Refer to the project ideas guidelines and Lotus Blossom worksheet as needed.

In addition to completing the tables below, you are to draw a **technical system sketch: a one-page sketch depicting a high-level technical system diagram of your proposed solution, such as a block diagram or dataflow diagram**. This should show both hardware and software modules.

The tables below are your initial proposal, and you may update these before your demonstration.

Functionality

Describe each of the basic functionalities required for the project in terms of your AV application. The functionality should be specific to the problem and user(s). Several functional statements are given in the example functional description for the Mars rover application in the Project Requirements document.

Basic Functionality	Mapping to AV Application
Cybot Communication	PuTTY
Cybot Movement	setWheels
Object Detection	PING and IR sensor
Object Avoidance	Bump and cliff sensors
Boundary Adherence	Cliff sensors
Arrival at Destination	Linear width of 4 thin pillars
User Interface	Python GUI
Base Station Control	PC
Other Application Specific Functionality (may be novel features for bonus points) Python custom GUI Music at arrival to destination	

Mapping to Platform

Briefly describe how each of the basic platform components required for the project will be used in your AV application.

Basic Platform Components	Usage in AV Application
Open Interface and iRobot Sensors	Cliff, IR, Bump, and PING sensors, move functions from open interface
Interrupts	Prioritize bumps and detect boundaries
ADC	Convert IR and PING data to digital values
Input Capture	Using UART to communicate through PuTTY
PWM	Rotate the servo
UART/WiFi	Send data to base station control
Other Platform Components or Modes (may be novel features for bonus points)	

Elements of the Test Field

Briefly describe a test field in the context of the real application (e.g., Martian terrain, city streets, etc.). Then state what each of the basic objects and other elements required for the test field represent in terms of the AV application. Draw and attach a **sketch of a possible simple test field for the lab**.

Test Field Description
Pictures below

Basic Objects and Other Elements	**Mapping to AV Application Test Field**
Tall objects (wide or composite)	Trees
Short objects	Rocks
Holes	Pits
Pillars (thin tall object)	Destination (door)
Out of bounds	Road
Destination zone	Front door
Other Application Specific Elements (may be novel features for bonus points or incidents to avoid)	

Serious Incidents to Avoid and/or Novel Features (Optional)

You may have identified novel features in the tables above. Enter them in the table below and propose possible bonus points if demonstrated successfully. In addition, describe any additional serious incidents that might happen in your test field for your AV application.

Novel Features	**Bonus Points**
Music plays when package arrives at front door.	+5

Serious Incidents	**Deductions**
Going out of bounds or bumping into pets or other obstacles	-1

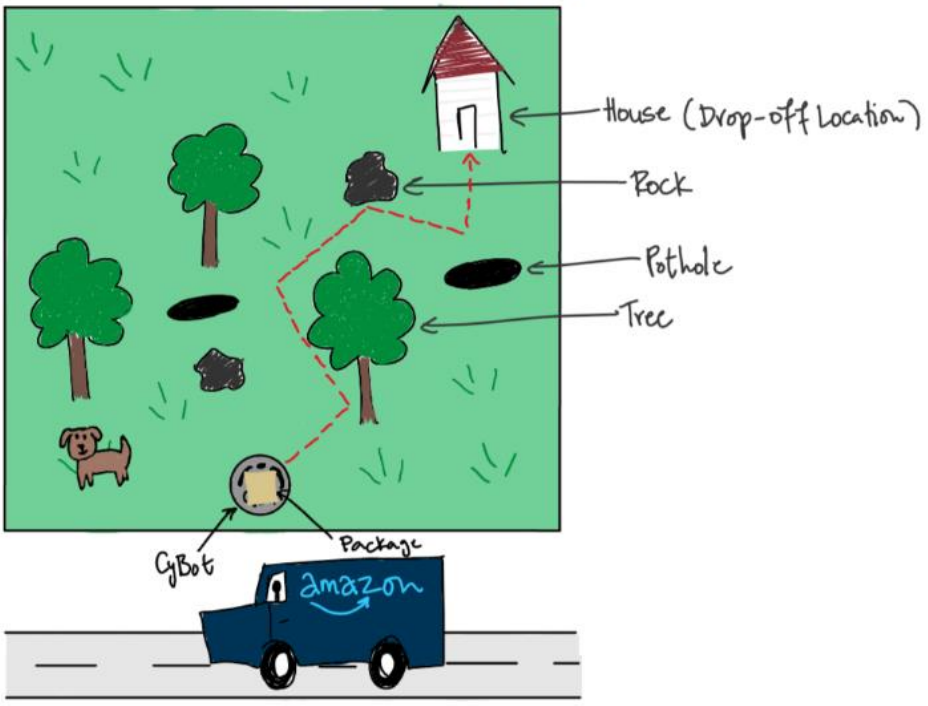
Sketches

Attach the following sketches to your submission. These were noted above in red.

- **Problem sketch**
- **Technical system sketch**
- **Test field sketch**

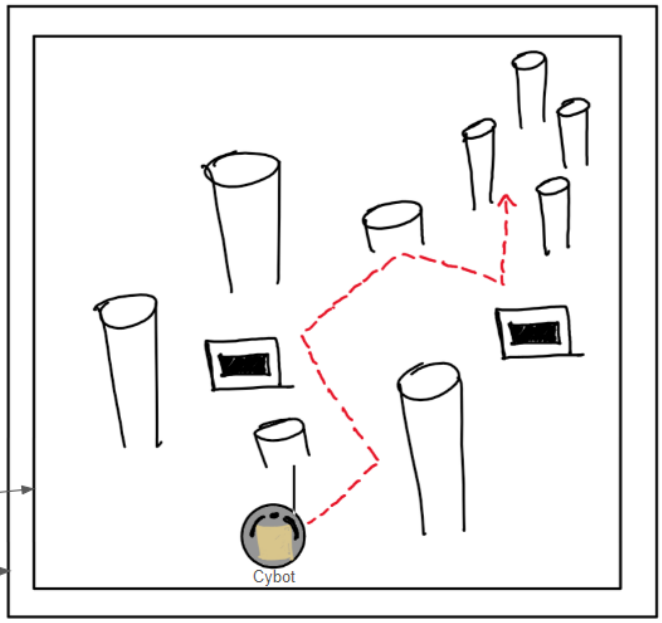
Visual communication is helpful for sharing information. There are many ways to represent your information. The problem sketch could be anything from a cartoon (informal) to a UML use case diagram (more formal). The technical system sketch is similar to what has been used/shown in class and lab. The test field sketch probably needs no explanation.

Problem Sketch



Test Field Sketch

Actual Object Sketch	Problem Sketch Object
Tall Objects	Trees
Short Objects	Rocks
Holes	Pits
Thin Pillars in square	The door (destination)
White Tape 1 (warning)	Sidewalk
White Tape 2 (out of bounds)	Road



Technical System Sketch

